



# Basic ESP Training

## Ampchart Analysis

# Ampchart Analysis

Assuming the recording ammeter is functioning properly, a number of changes in operating conditions can be defined by proper interpretation of the amp chart.

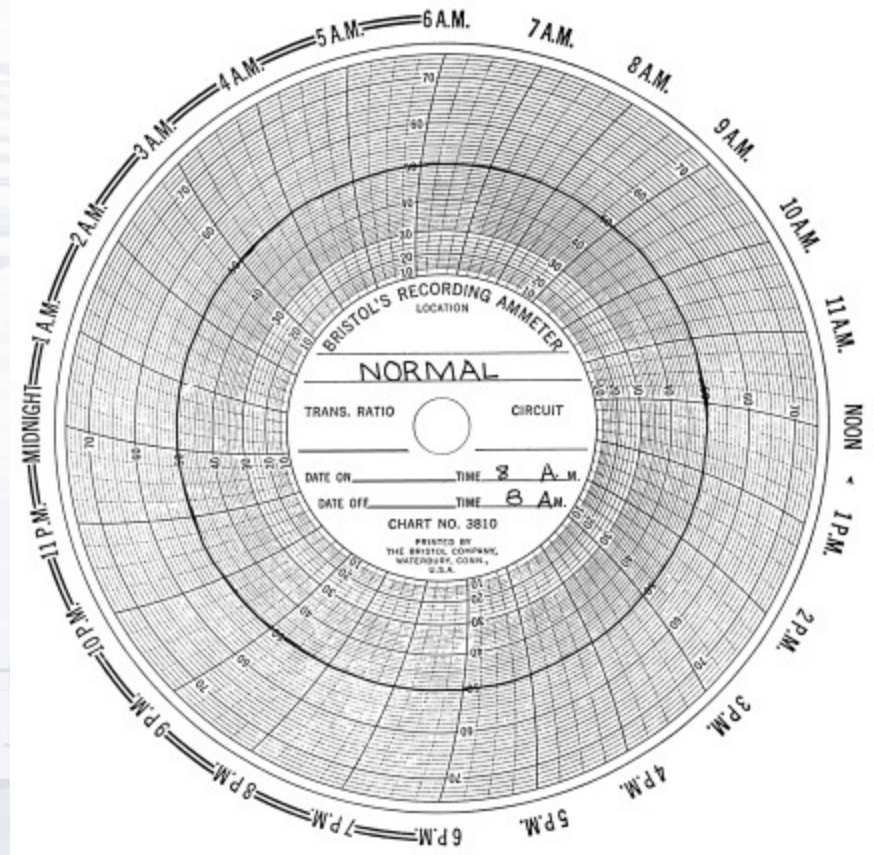
Some potentially damaging conditions are

- low-amperage, high-amperage or erratic-amperage operation
- primary power line voltage fluctuations

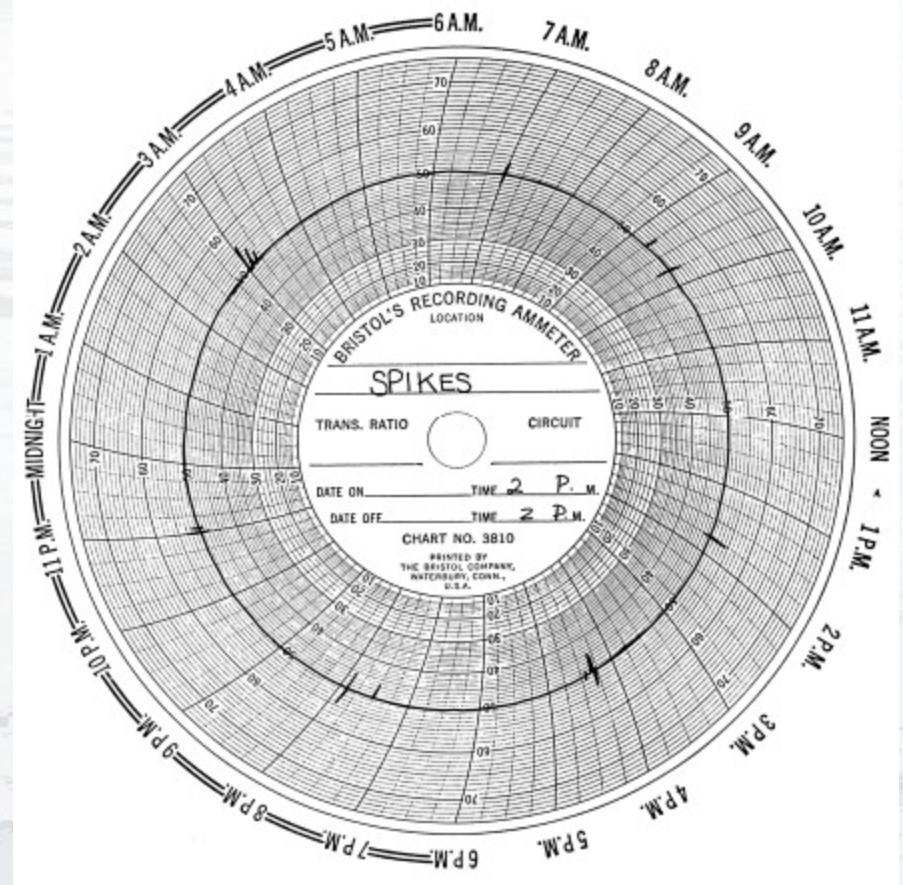
The following guidelines discuss the proper interpretation of ammeter charts and their interrelationship with other guides in the troubleshooting and preventive maintenance of submersible systems.

# Ampchart Analysis

- Normal Operation
- Smooth amp-curve
- Deviation from this normal curve may indicate potential problem or a change of well condition

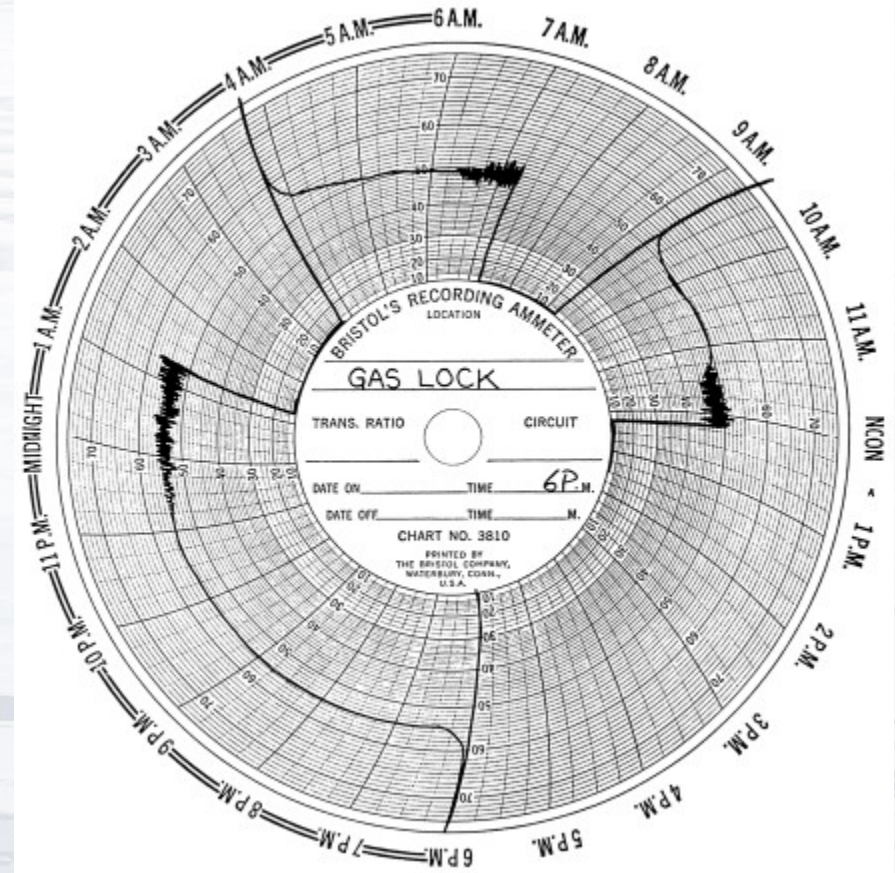


- Possible power fluctuation
- Amp curve relatively smooth, with the exception of several “spikes”
- Similar curve could be observed during natural disturbances such as lighting storm



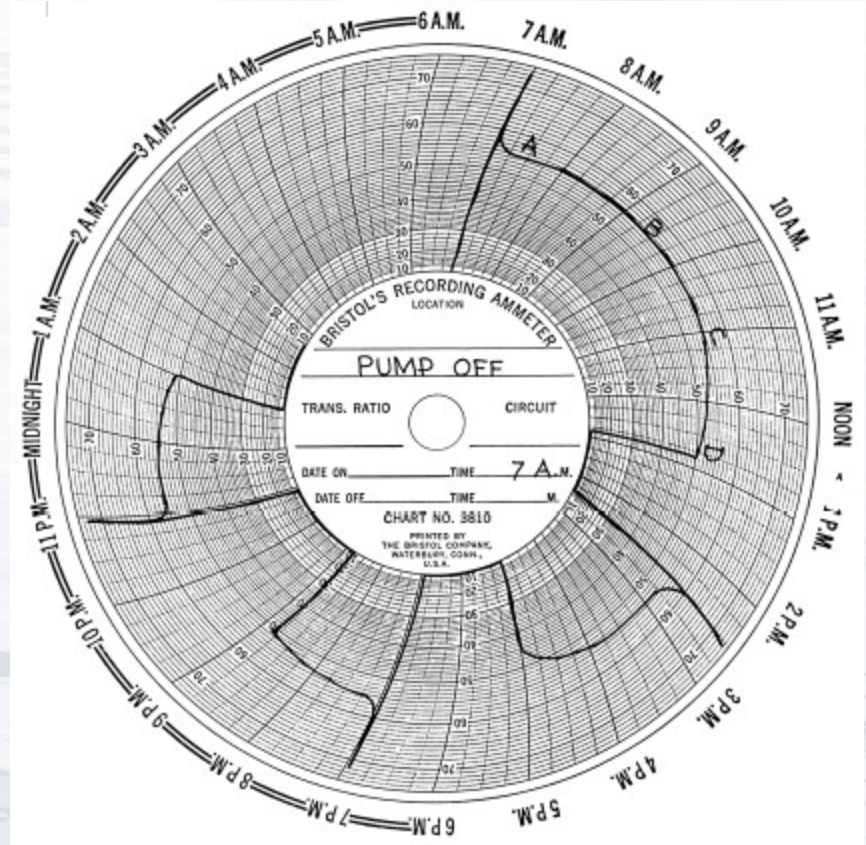
# Ampchart Analysis

- Possible Gas-lock
- Normal amp-curve observed during start up
- Followed by decline of current, and later shut down
- Could be remedied by changing the pump operating point



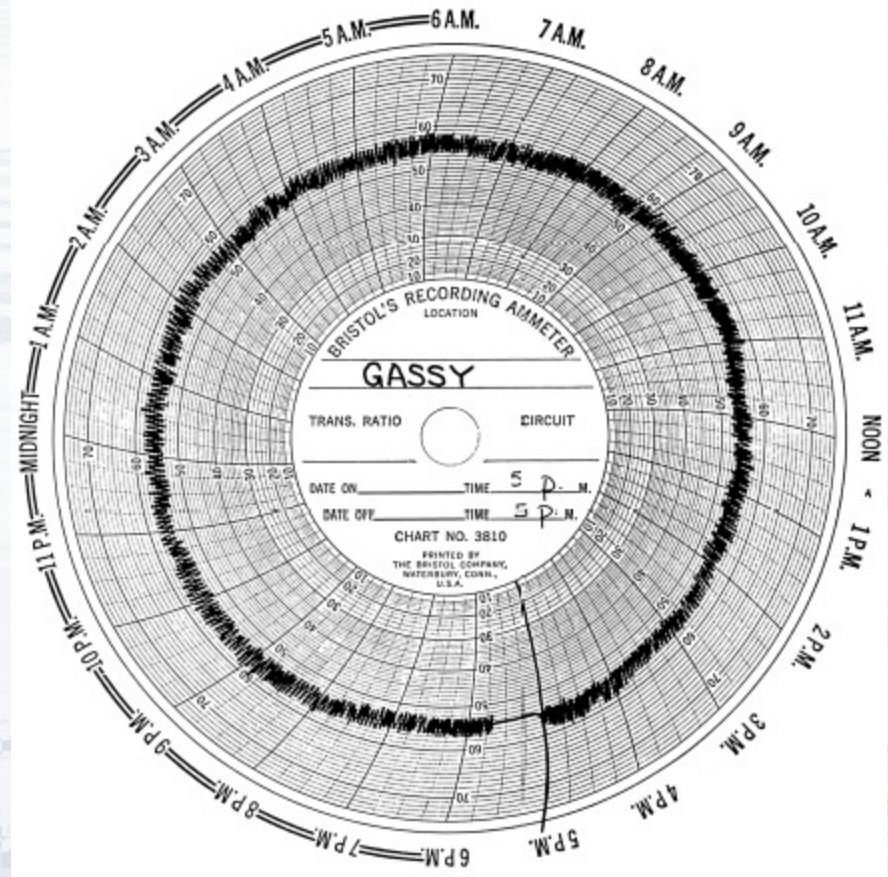
# Ampchart Analysis

- Possible fluid pump-off condition
- Similar to gas-lock curve, without the spikes
- Could be remedied by changing the pump operating point
- In the case pump already running at minimum point, may require pump change out



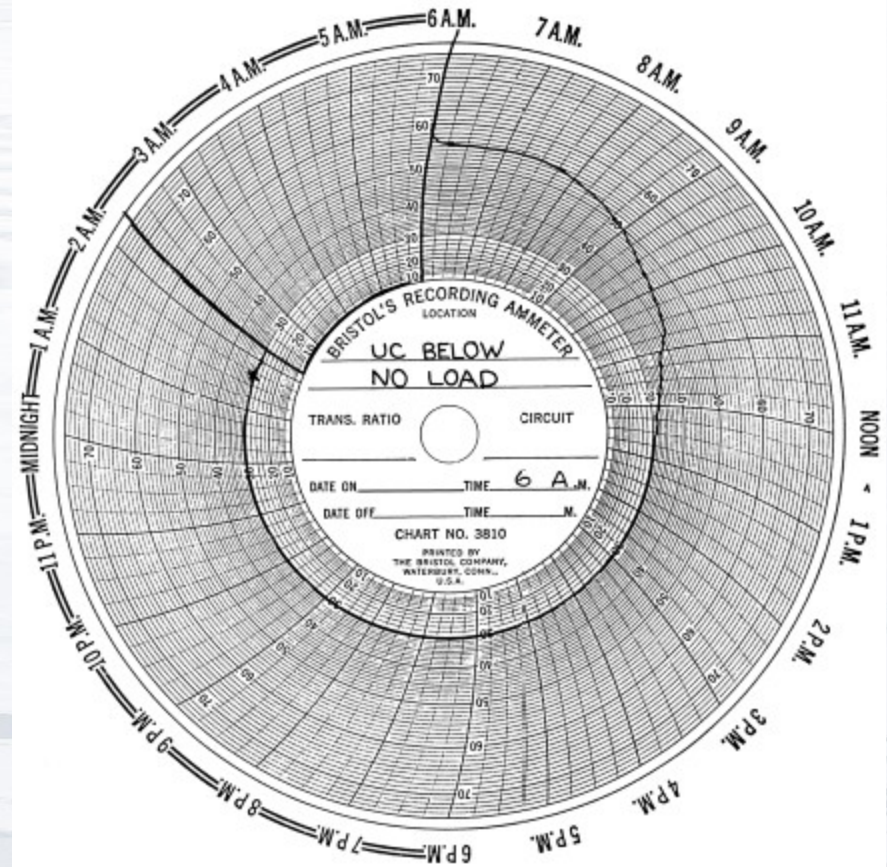
# Ampchart Analysis

- Gassy condition
- Happens as a result the ESP is pumping lighter fluid (gas)
- Normally, a drop of surface flow liquid flow rate could be observed
- Recommends to install proper gas handling device on subsequent installation



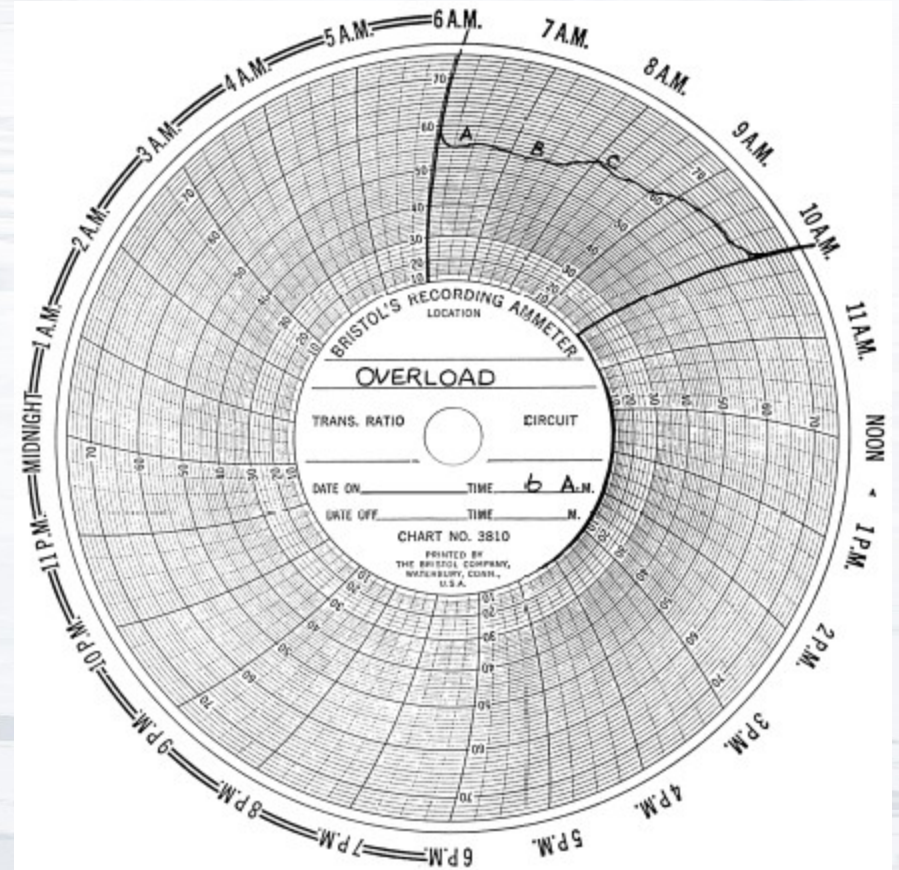
# Ampchart Analysis

- Underload condition
- Unit started normally, followed by decline in amperage
- Later, unit went to overload and tripped, quite possibly failed due to insufficient cooling of the motor
- Happened because unit UL relay was set below the “idle-amp”



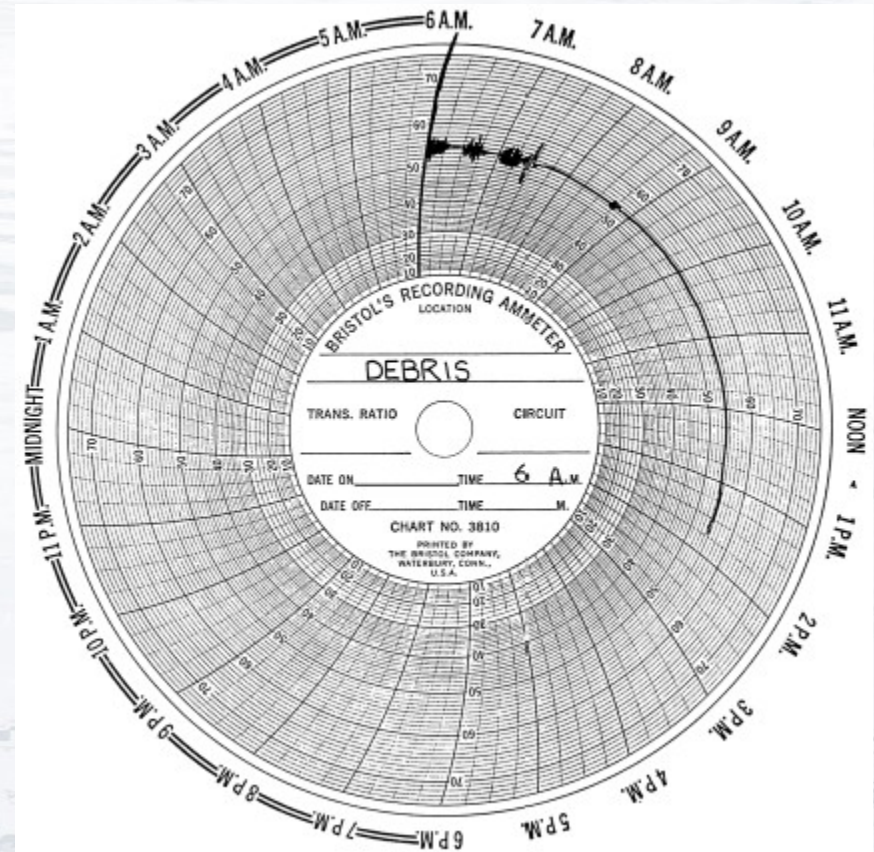
# Ampchart Analysis

- Overload condition
- Common causes of this type of shutdown are increases in fluid specific gravity or viscosity (such as heavy brines or muds), sand production, emulsions, or mechanical problems such as lightning, motor overheat or wearing equipment.
- Unit should not be restarted until problem could be identified and remedied.



# Ampchart Analysis

- Possible presence of debris
- Also possible that well is producing sand. In such case, it is recommended to hold back production rate and produce the well slowly until stabilize



# Ampchart Analysis

- Erratic behaviour
- This type of behavior is usually produced by fluctuations in fluid specific gravity or large changes in surface pressure
- Manual restart should not be attempted until the system is thoroughly checked by a service technician and the cause of the problem determined

